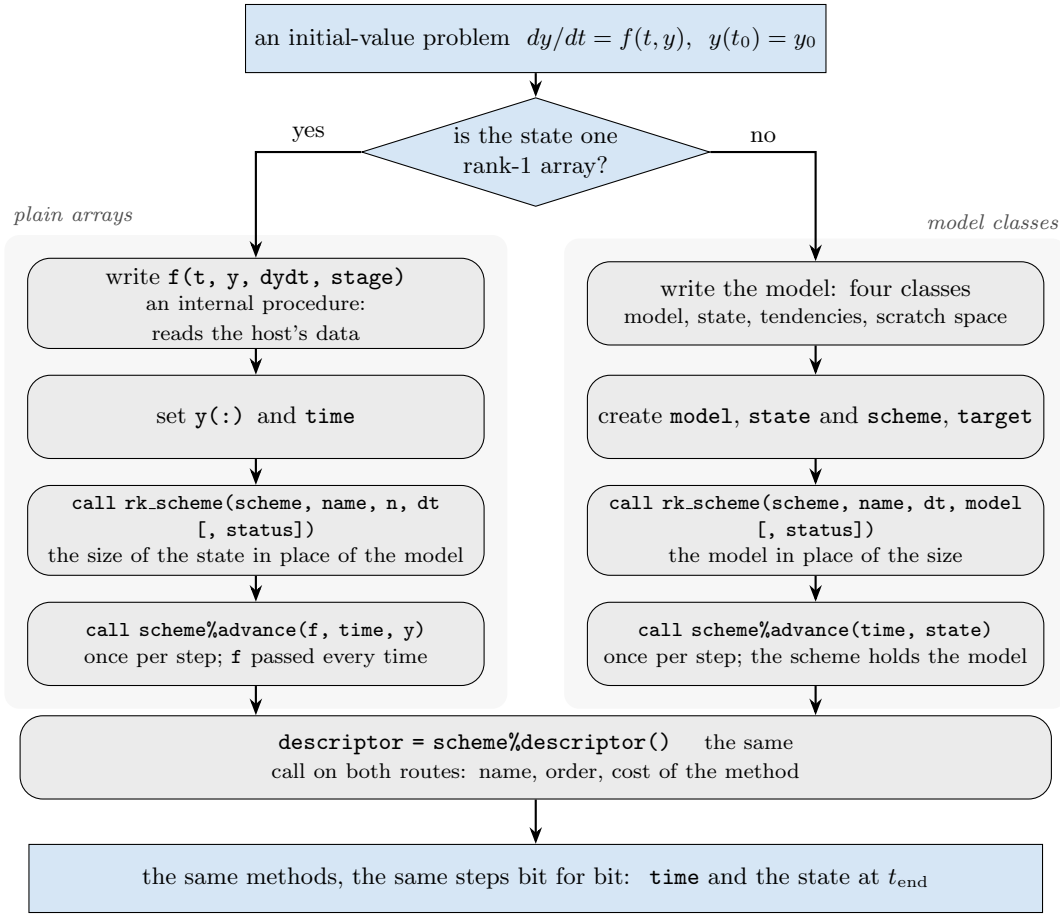

an initial-value problem $dy/dt = f(t, y)$, $y(t_0) = y_0$
is the state one rank-1 array?
yes
no
plain arrays
model classes
1
write f(t, y, dydt, stage)
an internal procedure:
reads the host's data
write the model: four classes
model, state, tendencies, scratch space
2
set y(:) and time
create model, state and scheme, target
3
call rk_scheme(scheme, name, n, dt [, status])
the size of the state in place of the model
call rk_scheme(scheme, name, dt, model [, status])
the model in place of the size
4
call scheme%advance(f, time, y)
once per step; f passed every time
call scheme%advance(time, state)
once per step; the scheme holds the model
5
descriptor = scheme%descriptor() the same
call on both routes: name, order, cost of the method
the same methods, the same steps bit for bit: time and the state at t_{end}